

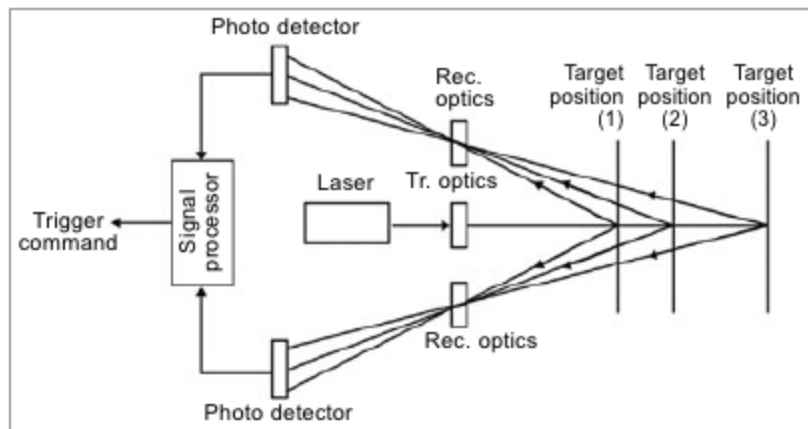


Fig. 6: Axially-symmetric arrangement of multiple aperture photo sensors



Fig. 7: Compact laser rangefinder module of DLEM family from Jenoptik



Fig. 8: LDS30M from Astech

sensors or laser proximity fuses. Triangulation technique is better suited for designing laser proximity sensors or fuses, as it can measure short distances down to a few metres with high accuracy. Range accuracy, however, falls off rapidly with increasing distance. In some cases, time-of-flight laser rangefinders have also been used to build laser proximity sensors.

Time-of-flight laser displacement or proximity sensors

Recent advances in laser technology have made possible the development of highly-miniaturised laser transmitters that can produce several tens of kilowatts of peak power in 1ns - 2ns laser pulses. These lasers are also capable of range update rates in excess of 500Hz, which is an essential requirement for proximity fuses used in artillery shells and aurally-dropped bombs.

An artillery shell or an aurally-dropped bomb has a travel velocity in the range of 200mps - 500mps in the terminal phase. At 500mps, it travels one metre in two milliseconds. This implies that, minimum range update rate to get a range resolution of one metre is 500Hz.

Also, miniaturised optics and high-speed APDs in SMD packages have led to the development of

compatible receivers. To add to this, these laser modules are available in different wavelengths, prominent among these being 905nm, 1064nm and 1550nm. One such compact laser rangefinder module of DLEM family from Jenoptik is shown in Fig. 7.

The latest addition to DLEM family, DLEM-20 weighs just 33 grams and measures ranges up to 5000m with an accuracy of 0.5m. It measures 50mm x 22mm x 34mm. DLEM-20 is ideally suited for handheld and weapon-mounted optronic systems. Its usage as of now as a proximity sensor or fuse may be limited due to maximum update rate of 25Hz.

Another laser rangefinder module for OEM applications that does not have the limitation of update rate is LDS30M from Astech (Fig. 8). LDS30M is an OEM laser module intended for distance measurements that require extremely fast data acquisition. It provides up to 30,000 range measurements per second.

Laser triangulation displacement or proximity sensors

Laser triangulation sensors for building proximity sensors or fuses are available as commercially-off-the-shelf (COTS) modules. These cover a wide range of performance specifications in terms of maximum measurable distance, measurement accuracy, range update rate and so on. Almost all of these contain a solid-state laser source, which is generally a semiconductor diode laser, and a position sensing detector (PSD) or CMOS/CCD detector.

A laser beam is projected on the target whose distance from the sensor or whose surface displacement is to be measured under extraneous stress. A fraction of laser beam power is reflected towards the sensor. Reflected radiation, whose intensity depends on reflectivity of the target surface, is focused onto the detector through focusing optics. As the target moves, the laser beam proportionally

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